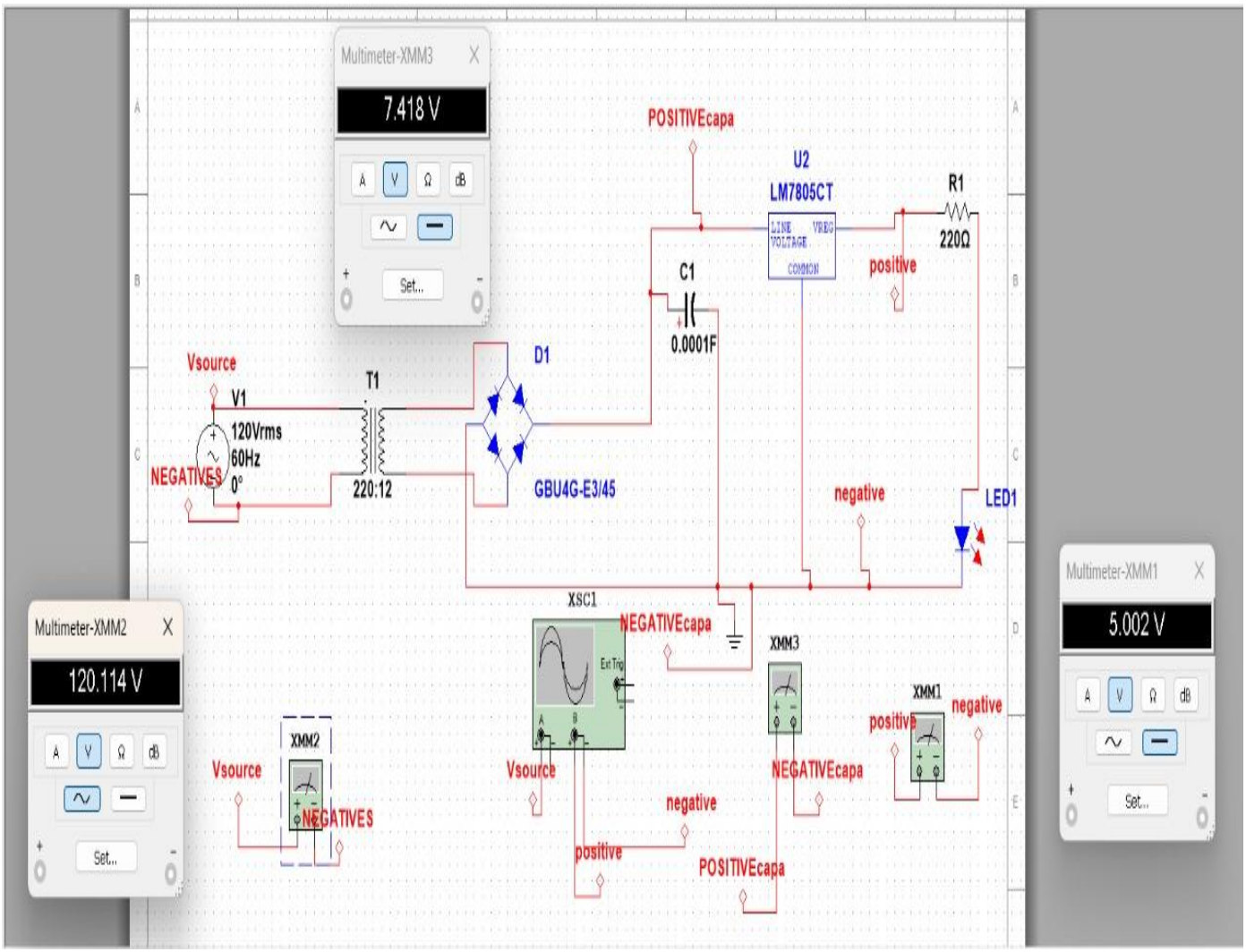
Circuit project

Analog and Digital Electronics

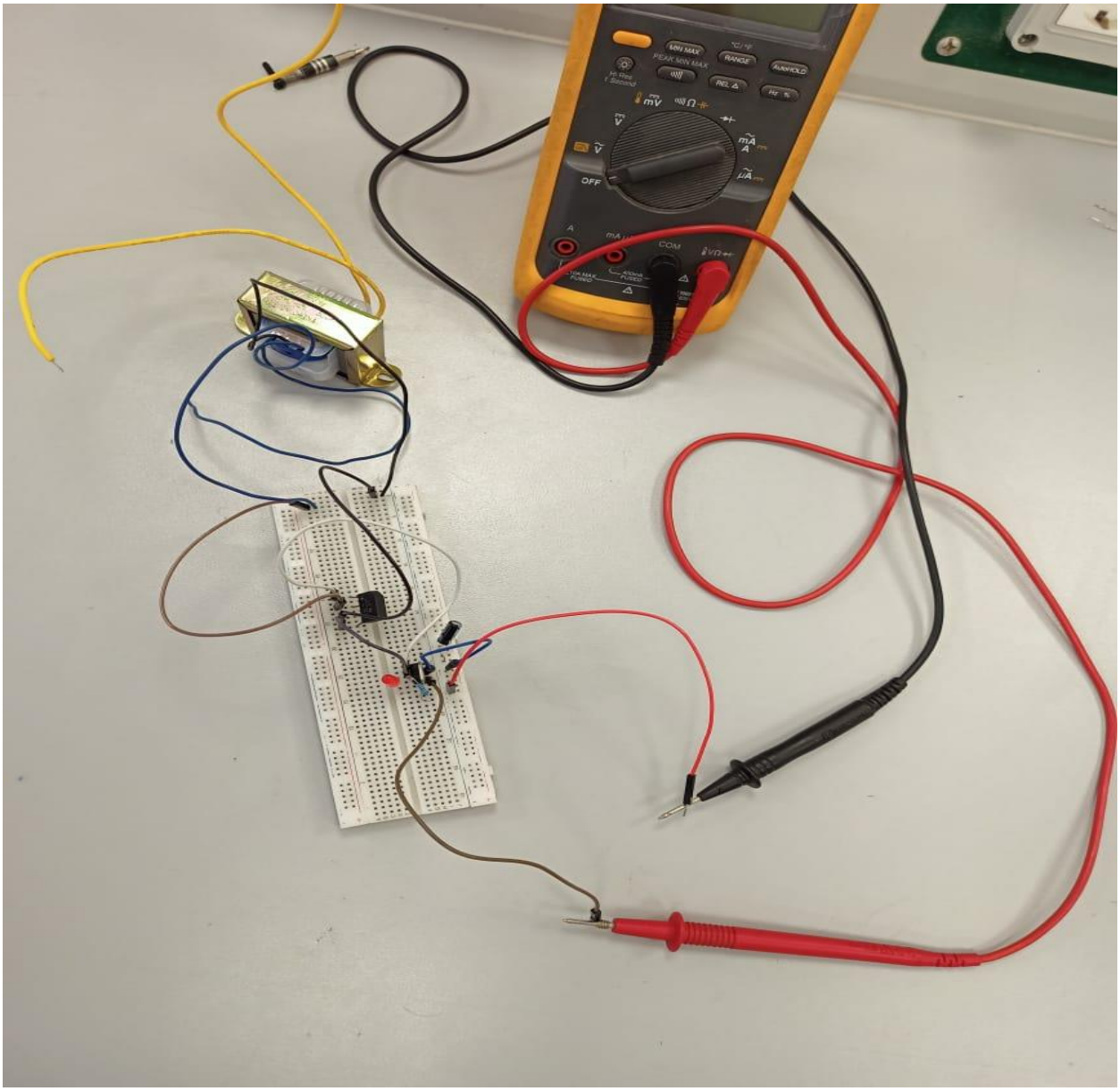
SPC 227

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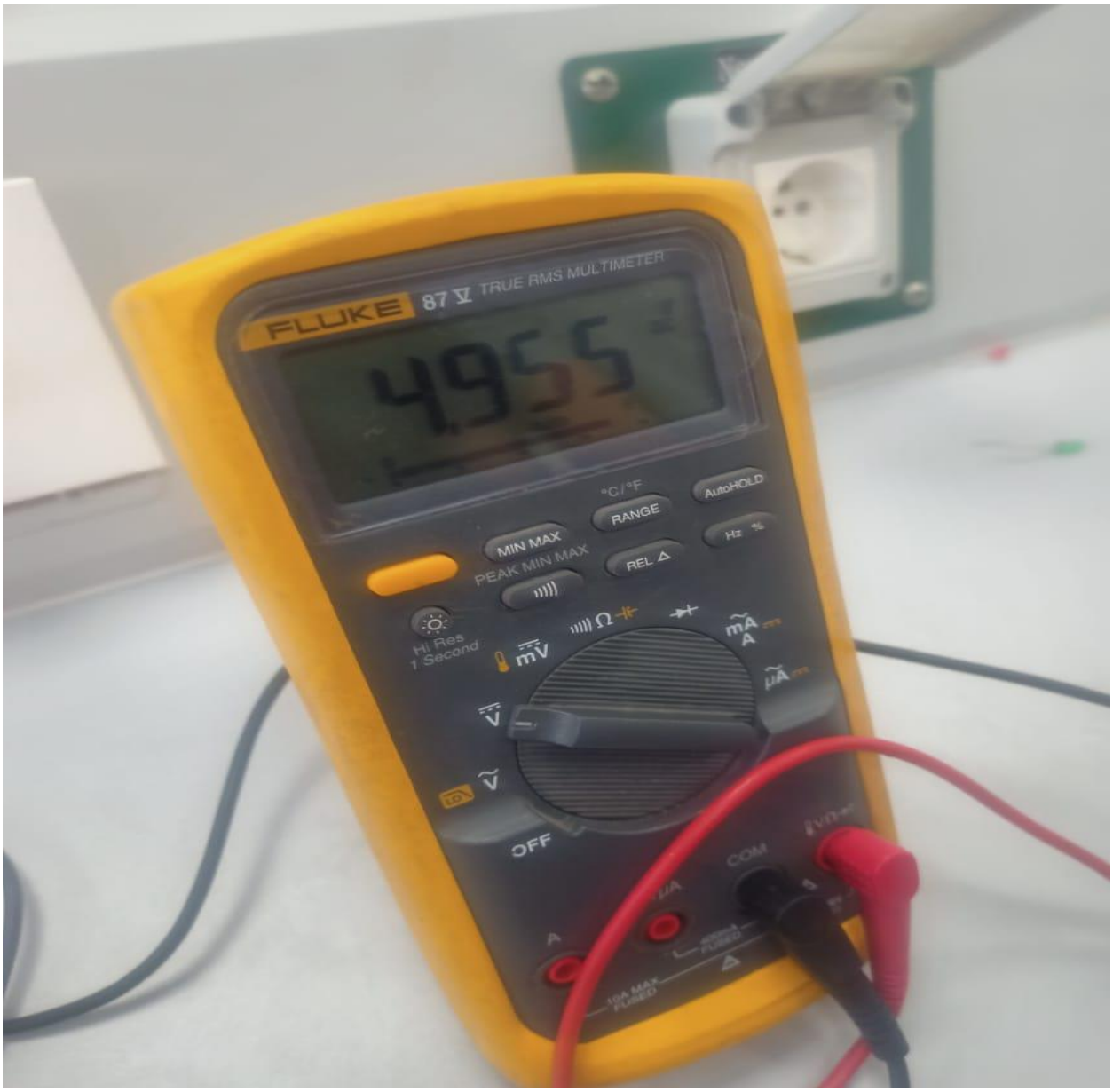
Simulation


Multimeter-XMM3
7.418 V
Set...
POSITIVEcapa
U2
LM7805CT
LINE VOLTAGE
VREG
COMMON
R1
220Ω
positive
C1
0.0001F
Vsource
V1
120Vrms
60Hz
0°
NEGATIVES
T1
220:12
D1
GBU4G-E3/45
negative
LED1
Multimeter-XMM1
5.002 V
Set...
Multimeter-XMM2
120.114 V
Set...
XSC1
Ext Trig
NEGATIVEcapa
XMM3
XMM1
positive
negative
XMM2
Vsource
NEGATIVES
Vsource
negative
positive
NEGATIVEcapa
POSITIVEcapa

Circuit



Result


FLUKE
87 V TRUE RMS MULTIMETER
4955
°C/°F
RANGE
AutoHOLD
MIN MAX
PEAK MIN MAX
REL Δ
Hz %
Hi Res
1 Second
mV
OFF
COM

Components

1. **Test board:** It provides a platform to test multiple electrical components like resistors, capacitors, inductors, transistors, and integrated circuits.
2. **Transformer:** it transfers electrical energy between two or more circuits through electromagnetic induction. It is used to step up or down the levels of voltage in alternating currents electrical circuits. Moreover, it provides isolation between circuits to assure safety and prevents ground loops and interference between circuits.
3. **Capacitor:** its main function is storing and releasing electrical energy in a circuit. Thet are also used in power supply circuits for filtering and signal filtering.
4. **Regulator:** it is used to maintain a constant output voltage or current level. It also ensures stable operation of electronic devices and systems. It plays main role in voltage regulation, current regulation, and protection.
5. **Resistor:** it is an electric component that regulates and limits the flow of electrical current in circuits. They play fundamental role in current limiting, voltage division, signal conditioning, biasing, and pull up and down resistors.
6. **Plug:** it is like a connection between electrical device and power supply. It provides electrical connection, contact points, mechanical support, polarization, safety features, and compatibility.